Heat Transfer to a Fluid in a Stirred Tank

Final Lab Report

Unit Operations Lab 2

February 16, 2026

Group 3, Tuesday 9 AM

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Table of Contents

1. Executive Summary

This experiment evaluated heat transfer performance in a steam-jacketed stirred tank by determining the inner heat transfer coefficient (hi) and overall heat transfer coefficient (Ui) under baffled and unbaffled conditions. The objective was to quantify how impeller speed and flow behavior influence convective heat transfer and compare experimental results with established heat transfer correlations.

Heat was supplied to the vessel through condensing steam in the jacket, while heat was removed using an internal cooling coil and an external recirculation heat exchanger. An energy balance an the stirred tank fluid was used to calculate the heat transfer rate (Q13). Experiments were conducted at multiple impeller speeds while maintaining the bulk fluid temperature between 65°C and 85°C. Trials were performed both without baffles and with four vertical baffles inserted to evaluate the effect of mixing enhancement.

Results showed that increasing impeller speed increased both hi and Ui due to improved turbulence and reduced thermal boundary layer thickness. The addition of baffles significantly enhanced heat transfer by suppressing vortex formation and promoting radial mixing. Experimental values of hi followed the expected dependence on impeller Reynolds number, though deviations from the literature correlation were observed due to geometric differences and experimental assumptions. Overall, the results demonstrate that agitation intensity and vessel configuration strongly influence heat transfer performance in stirred tank systems.

2. Introduction

The objective of this experiment is to evaluate heat transfer in a jacketed stirred tank by experimentally determining the inner film heat transfer coefficient (hi) and overall heat transfer coefficient (Ui), and to examine the effects of impeller speed and baffle insertion on heat transfer performance. Heat transfer in stirred vessels is a critical design parameter in many chemical and industrial processes, including continuous stirred tank reactors (CSTRs), mixing operations, crystallization, and temperature control of exothermic reactions.

In jacketed vessels, heat is transferred from condensing steam through the vessel wall and into the agitated fluid. The rate of heat transfer depends strongly on fluid flow patterns, mixing intensity, and thermal properties. Convective heat transfer coefficients are often correlated using dimensionless groups such as Reynolds, Prandtl, and Nusselt numbers. However, due to the complex flow fields in stirred tanks, experimental evaluation is necessary to obtain reliable design data.

The apparatus used in this experiment consists of a steam-jacketed stirred vessel equipped with an internal cooling coil and an external recirculation heat exchanger. The impeller speed was varied using a variable power supply, and baffles were inserted or removed to modify the flow pattern within the tank. Temperatures and flowrates were measured to perform steady-state energy balances, from which heat transfer rates and coefficients were calculated. The experimentally determined values of hi were compared to empirical correlations to assess agreement with established heat transfer theory.

3. Theory

Heat transfer in a jacketed stirred tank occurs from condensing steam in the jacket to the liquid inside the vessel. Under steady-state conditions, the heat transfer rate is the same through each layer because no energy accumulates in the wall. Heat is transferred from the steam to the outer wall surface during condensation, then through the solid wall by conduction, and finally from the inner wall surface to the liquid by convection (6). Conduction through the wall follows Fourier's law (Equation 1, Table 1), which states that the heat transfer rate is proportional to the thermal conductivity of the material and the temperature gradient across it (9).

Because each interface and solid phase presents a different resistance to heat transfer, the temperature profile across the tank wall appears as shown in Figure 1 below.

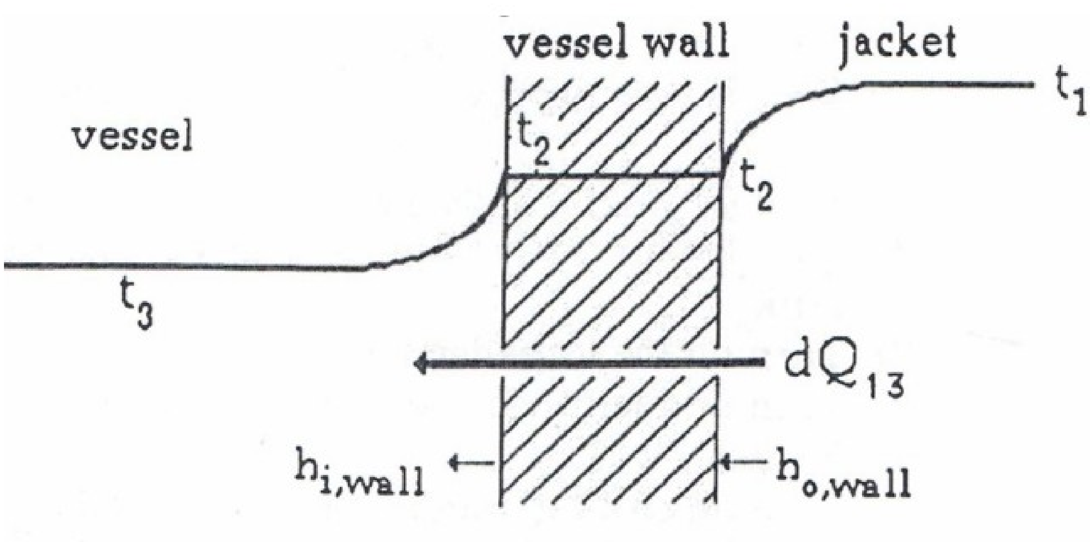


Figure 1: *Heat Transfer Across the Tank Wall (6)*

Most of the temperature drop occurs in the thin fluid boundary layers on either side of the wall. The heat transfer across the outer and inner surfaces can be described using convection equations (Equation 2, Table 1). These equations relate the heat transfer rate to the heat transfer coefficient and the temperature difference between the wall and the fluid. Combining these resistances gives the overall heat transfer equation (Equation 3, Table 1). The overall heat transfer coefficient is defined by Equation 4 in Table 1, which shows that the total resistance is the sum of the steam side and liquid side resistances (6).

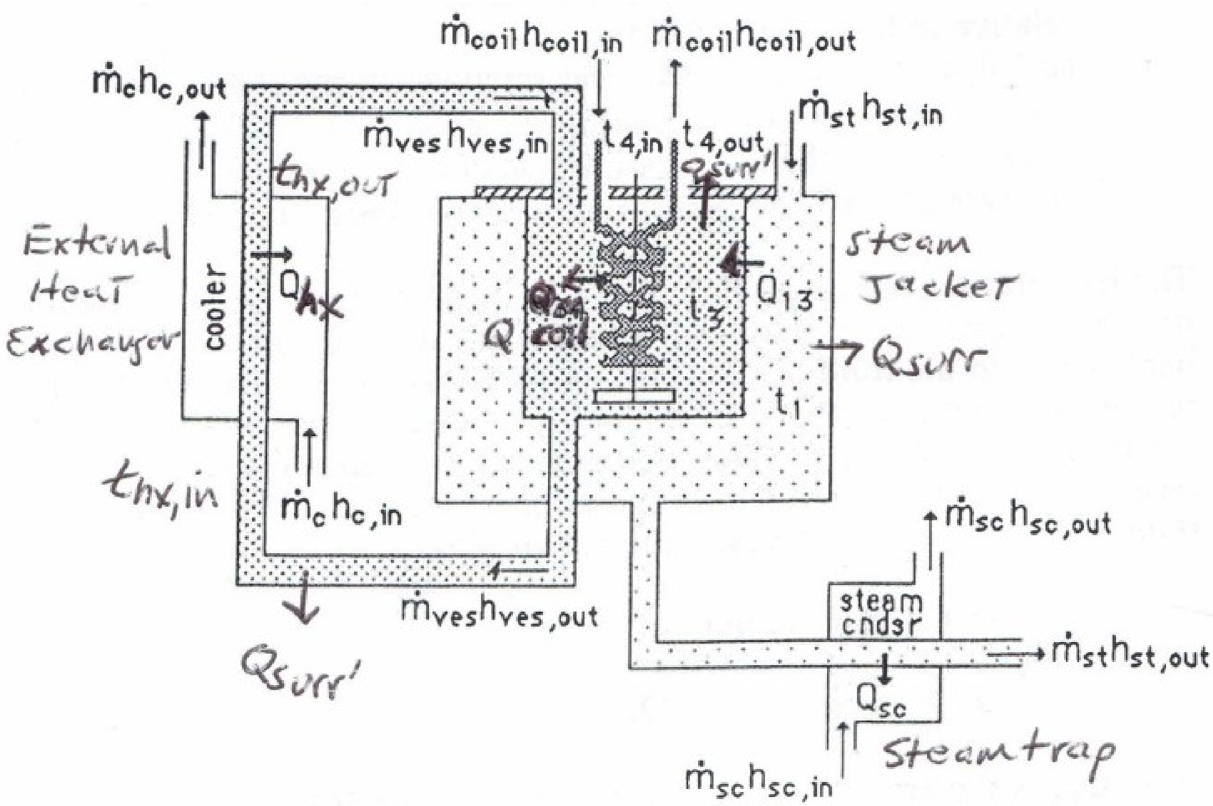


Figure 2: *Mass and Energy Flow in Stirred Tank System (6)*

The apparatus used in this experiment includes a steam jacket, an internal cooling coil, and an external heat exchanger. The mass and energy flow in the system is shown in Figure 2.

At steady state, the heat added by the steam condensation must equal the heat removed by the cooling coil, the external heat exchanger, and any heat lost to the surroundings. This follows directly from the first law of thermodynamics for a steady-flow control volume (7). The heat removed can be calculated using Equation 5 in Table 1, which is the steady state energy balance for the experimental system (Lab Manual).

The liquid side heat transfer depends strongly on mixing conditions inside the tank. Because the flow pattern generated by the impeller is too complex, heat transfer is described using dimensional analysis. The Nusselt number (Equation 8, Table 1) represents the dimensionless heat transfer coefficient, while the impeller Reynolds number (Equation 9, Table 1) represents the intensity of mixing. The Prandtl number (Equation 10, Table 1) accounts for fluid thermal properties (6). Increasing impeller speed increases the Reynolds number, enhances turbulence, and increases the liquid side heat transfer coefficient (10). The addition of baffles prevents vortex formation and improves mixing, which generally increases heat transfer compared to an unbaffled tank (8).

The data collected in this experiment includes temperatures at multiple points in the system, flow rates of the cooling water and recirculation streams, impeller speed, and the mass of steam condensate collected over time. This data will be used to determine the overall and individual heat transfer coefficients for the jacketed stirred tank by using the equations shown in Table 1.

Table 1: Relevant Equations for Heat Transfer Experiment		
Title	**Equation**	**Components**
1. Fourier's Law Conduction Through Wall		• = heat transfer rate (W) = thermal conductivity (W/m·K) • = area (m²) • = temperature gradient
2. Convective Heat Transfer at Interfaces		• , = heat transfer coefficients (W/m²·K) • , = areas • = steam temp • = wall temp • = bulk liquid temp
3. Overall Heat Transfer Equation		• U = overall heat transfer coefficient (W/m²·K) • A = heat transfer area (m²)
4. Thermal Resistances in Series		• = liquid-side coefficient • = steam-side coefficient
5. Overall Steady-State Energy Balance		• = heat from steam; = heat removed in HX • = heat removed in cooling coil • = heat loss
6. Heat Removed by External Heat Exchanger		• = mass flow rate (kg/s) • = specific heat (J/kg·K) • = tank temp • = HX outlet temp
7. Heat Removed by Cooling Coil		• = coil water flow rate (kg/s) • = specific heat • = coil outlet and inlet temps
8. Nusselt Number		• h = heat transfer coefficient • = tank diameter

		• k = thermal conductivity
9. Impeller Reynolds Number		• ρ = density (kg/m^3) • = impeller diameter (m) • n = rotational speed • mu = viscosity
10. Prandtl Number		• = specific heat • mu = viscosity • k = thermal conductivity
11. Empirical Correlation for Unbaffled Stirred Tank		• = tank Nusselt number; valid for unbaffled stirred tank

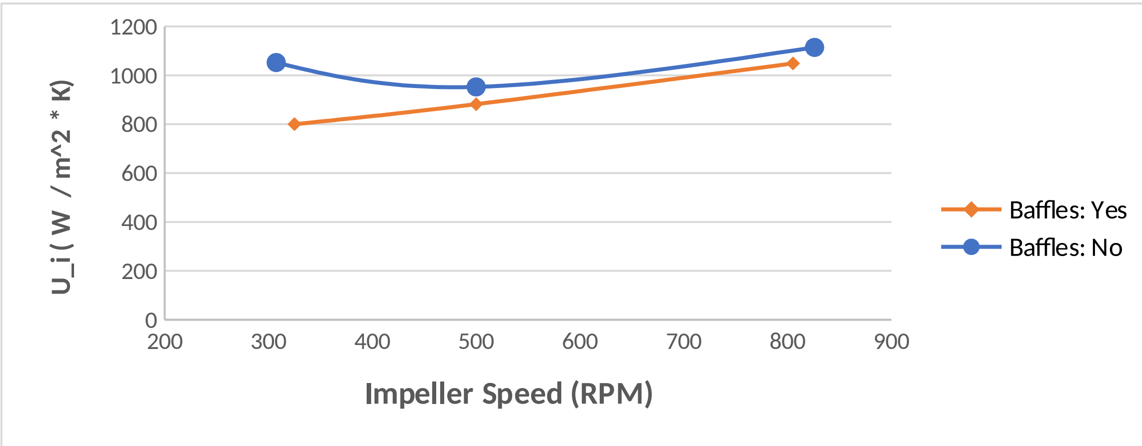
4. Results

Both experiments 1 and 2 in this lab focused on finding the heat transfer (quantified by several heat transfer coefficients) of the system using a set of given process variables. Impeller speed and baffle placement (with/without baffles) were varied across trials. Additionally, cooling water and recirculation flow rate through the heat exchanger were adjusted to maintain a tank temperature between 65°C - 85°C.

With our calculations, we found that Q_{13} increases with impeller speed whether baffles were included. Q_{13} were in the range of 7468 - 10599 W with the cooling coil removing most of the heat.

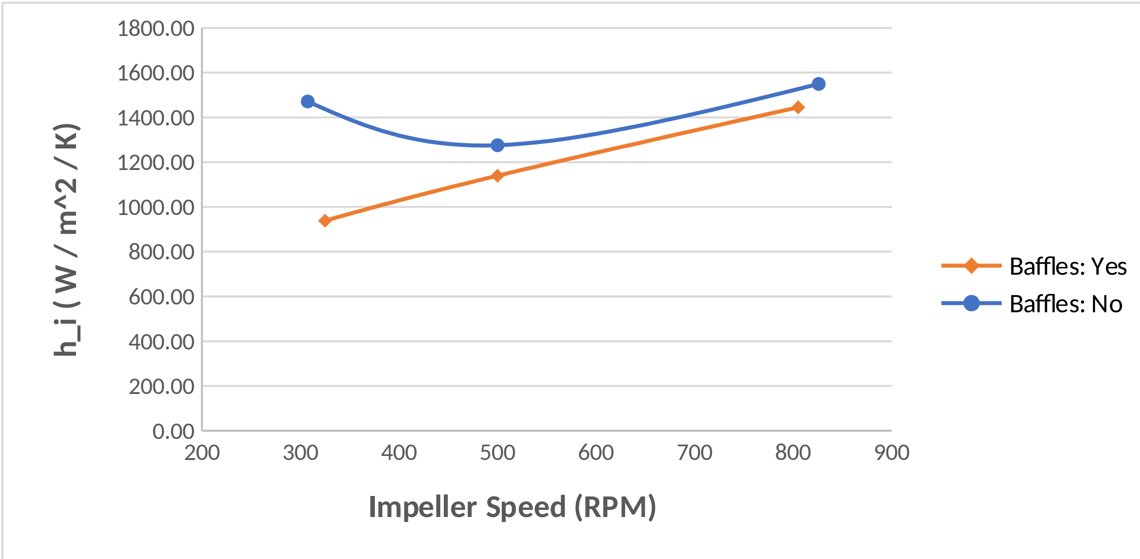
These values can be extrapolated into metrics that are more easily compared, namely the overall heat transfer coefficient (U), outer heat transfer coefficient (h_o), and heat transfer coefficient (h_i). Each of these coefficients can be plotted for the range of impeller speeds and baffled/unbaffled status as shown below:

From our experiments, we saw as the impeller speeds increased, the overall heat transfer coefficient, U_i, increased in both cases with the inclusion of the baffles. This makes sense that with higher mixing, the rate of heat transfer would increase. However, with Trial 2, with no baffles at 500 RPM, the tank spilled over with steam, which shows the irregularity with the large U_i.



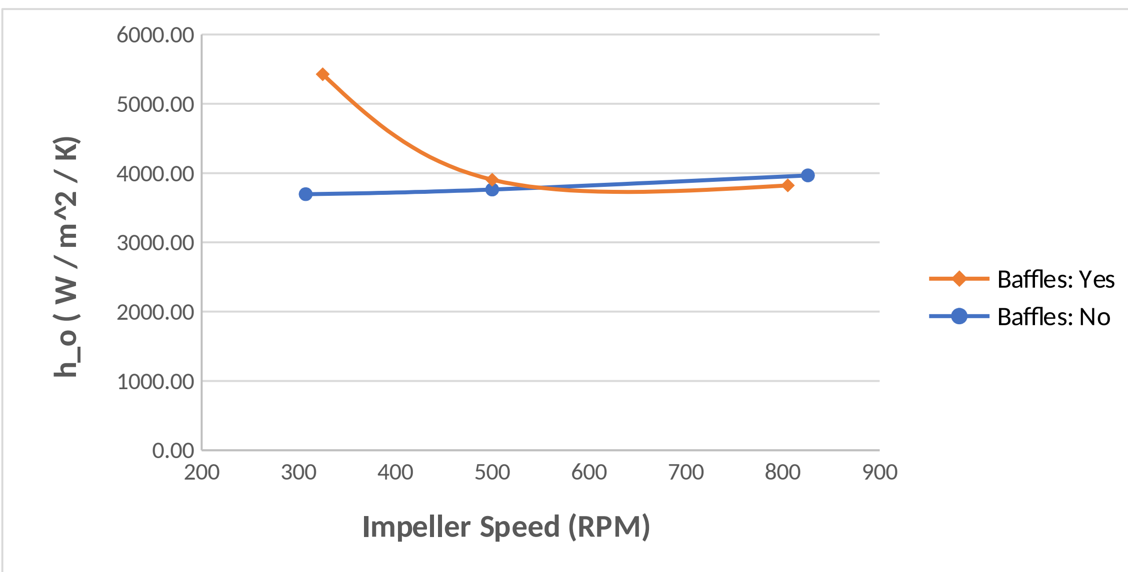
Graph 1 – U_i vs Impeller Speed

Like U_i, the heat transfer coefficient, h_i. followed a similar trend as U_i. As h_i is used to calculate U_i, the issue with Trial 2, shows the irregularity. With the use of baffles, h_i increased at a constant rate which would make sense as baffles improve mixing.



Graph 2 – h_i vs Impeller Speed

The other heat transfer coefficient, h_o, during Trial 4, the lowest RPM and with the use of a baffle showed an outlier with the low condensation gathered.



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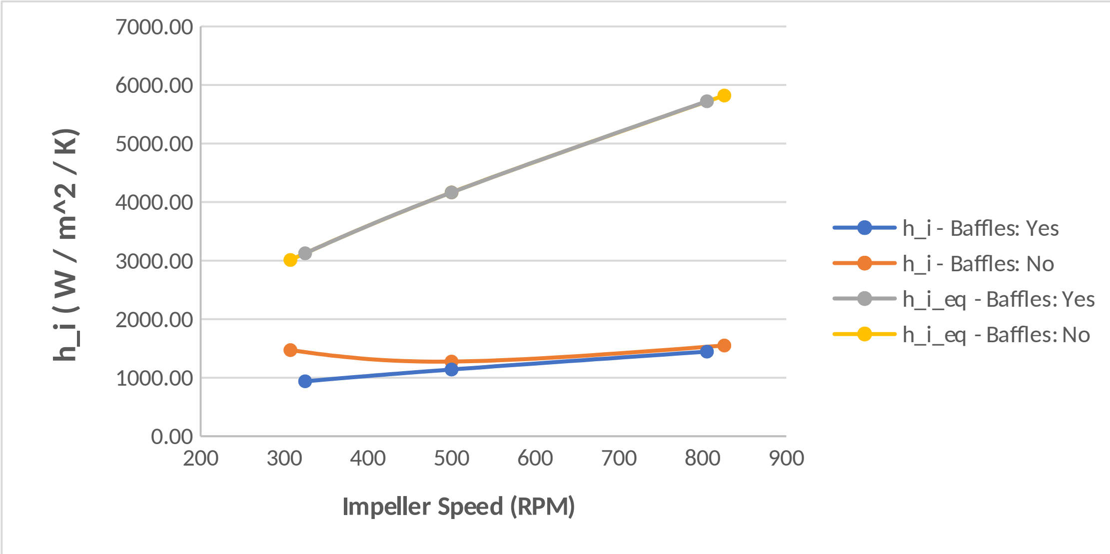
Graph 3 – h_o vs Impeller Speed



Graph 4 - h_i_eq vs h_i

However, with our comparisons of the calculated and experimentally found h_i, showed a significant % difference. Even with this, the trend with the increase of the impeller speed led to an increase of the heat transfer coefficients.

5. Discussion

Generally, the observed behavior of the stir tank followed theoretical expectations as impeller speed increases, although there was more significant deviation at the lowest impeller speed in the unbaffled trial. This is shown by *Graph 1 and* 5, where the heat transfer coefficients increase at greater impeller speed. Additionally, the baffled vs. unbaffled trials exhibited an unexpected trend, as the unbaffled trials have consistently higher U_i and h_i values than the baffled trials. According to the literature [cite lab manual], the opposite is expected as the baffles increase the mixing within the tank, leading to greater heat transfer. Despite this, the relative magnitude of each of the coefficients is still consistent with expected values, with U_i between 620-860 $W*m^{-2*}K^{-1}$, h_i between 700-1100 $W*m^{-2*}K^{-1}$, and h_o between 3677-5398 $W*m^{-2*}K^{-1}$, as the majority of the resistance to heat transfer is within the process fluid itself [cite lab manual]. The assumption that there is negligible heat-transfer-resistance by the wall is further validated by the relative sensitivity of h_i vs. h_o to changes in impeller speed, as shown by the steeper slope of *Graph* 6 *as opposed to Graph* 7 Sensitivity can also be compared between the between the baffled and unbaffled cases in the same figures, which shows a very small difference between these two experimental conditions (although this may not be accurate, given the inconsistency with the theory discussed above).

With a limited amount of data collected in the experiment it is difficult to determine what component of the error can be attributed to normal fluctuation in the experimental conditions due to the sensitivity of the impeller speed, amongst other factors; potential experimental error caused by not waiting for the system to reach steady state or the formation of vortexes in the mixer; and expected deviations from theoretical values due to discrepancies in the experimental setup, such as using an impeller rather than a paddle mixer. Graph 8 suggests that a significant portion of this error may be due to a mismatch between the theoretical and actual setup, as the values for h_i in all cases are consistently much lower than the experimental values. One assumption used to develop the theory in the lab manual [] is that there is minimal unrecorded heat loss from the system. This may not be accurate, as with some trials there was some temperature instability and minor leakage of both liquid and steam leaks from the tank.

Although a clear connection between both agitation intensity and tank configuration to heat transfer is demonstrated by the findings, further conclusions cannot easily be drawn without more rigorous testing due to the inconsistencies in the dataset. In addition to testing more intermediate impeller speeds, varying the feed of both the cooling water and recirculation flow may allow for a greater understanding of the system.

6. References

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9. Appendix I: Data Tabulation/Graphs

Raw Data:

The calculated Q_{13} values (see **Appendix I** for calculations) are as follow:

Calculated Q_{13} values							
Trials	**Baffles (Y/N)**	**Impeller Speed (RPM)**	**dt_coil (°C)**	**Q_coil**	**dt_hx**	**Q_hx (W)**	**Q_13 (W)**
1	N	307.4	29	7168	21	300	7468
2	N	500	33	8271	37	539	8810
3	N	826	39	9650	45	655	10304
4	Y	325	29	7306	54	782	8088
5	Y	500	33	8133	54	782	8916
6	Y	805.2	39	9787	56	811	10599

Table 2 - Calculated Heat Transfer Rates

Trials	**Baffles (Y/N)**	**Impeller Speed (RPM)**	**dT_13 (K)**	**U_i (W / m^2 * K)**	**H_o(W / m^2 / K)**	**H_i (W / m^2 / K)**	**h_i_eq(W / m^2 / K)**	**% diff**
1	N	307.4	18	1052	3677	1474	2931	49.69
2	N	500	24	953	3743	1278	4053	68.47
3	N	826	24	1114	3946	1553	5664	72.59
4	Y	325	26	800	5398	940	3041	69.11
5	Y	500	26	882	3884	1141	4053	71.84
6	Y	805.2	26	1049	3803	1448	5569	74.00

Table 3 - Calculated Heat Transfer Coefficients

Cooling Coil Flowrate (L / s)	Recirculation Flowrate (L / s)	T1 (°C)	T2 (°C)	T3 (°C)	T4 (°C)	T5 (°C)	T6 (°C)	T7 (°C)
0.0593046	0.00346995	10	82	39	58	61	73	51
0.0593046	0.00346995	10	76	43	32	39	64	50
0.0593046	0.00346995	10	76	49	28	31	70	51
0.0593046	0.00346995	9.4	74	39	22	20	172	50
0.0593046	0.00346995	9.4	74	42	19	20	42	49
0.0593046	0.00346995	9.4	74	49	18	18	51	50

Table 4 - Raw Data

Condensate V (mL)	Time (s)	Rate (L/s)
140	7.94	0.017632242
130	7.78	0.016709512
152	10.66	0.014258912
85	15.26	0.005570118
140	9.36	0.014957265
138	8.66	0.015935335

Table 5- Raw Data

m_s (kg / s)	Γ (kg / (s*m))	Re_f	h_o (W / m^2 / K)	h_i (W / m^2 / K)	Re_imp	P_γ	N_u_T	h_i_eq(W / m^2 / K)	% diff
0.01759	0.06	373.42	3676.69	1474.33	85218.11	4.184	1123.37	2930.51811	49.69
0.01667	0.05	353.88	3743.15	1277.95	138611.11	4.184	1553.69	4053.111056	68.47
0.01423	0.05	301.98	3946.36	1552.77	228985.56	4.184	2171.23	5664.068961	72.59
0.00555	0.02	117.97	5398.47	939.51	90097.22	4.184	1165.84	3041.333704	69.11
0.01492	0.05	316.77	3883.96	1141.26	138611.11	4.184	1553.69	4053.111056	71.84
0.01590	0.05	337.	3802.	1447.	223219	4.1	2134.	5568.578	74.

	48	82	84	.33	84	62	483	00

Table 6- Raw Data

10. Appendix II: Error Analysis

The experimental inner heat transfer coefficients differed significantly from the values predicted using the empirical Nusselt correlation with percentage differences ranging from 49.69% to 74%. While the expected trend of increasing heat transfer coefficient with impeller speed was observed, the magnitude of of deviation indicates the presence of systematic and random experimental error.

Sources of error may include the assumption of negligible heat loss to the surroundings, uncertainty in the condensate collection measurements and the assumption that the system operates at steady state. If sufficient time was not allowed for the tank temperature to stabilize this would introduce error into the energy balances

% diff
49.69
68.47
72.59
69.11
71.84
74.00

11. Appendix III: Sample Calculations

Calculation of offset for experiment :

10. Appendix IV: Individual Team Contributions

Table 7: *Individual Team Contributions*

Contribution	**Team Member**
Executive Summary, Introduction	Hiba Naqvi
Theory	Angelica Ramirez
Results	Noah Guale
Discussion	Basil Hashim
Appendix I, II	Jazmin Aca
References	All